

Twitter Sentiment Analysis

Apple & Google Products

NLP Classification + Neural Networks + Model Interpretability

Author

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Step 1: Business Understanding & Data Overview

Business Questions

- Q 1** Apple & Google receive thousands of tweets daily — how can brand teams manually read and categorise all this data?
- Q 2** What can eliminate delays in responding to negative brand signals caused by feature failures or software updates?
- Q 3** How do we measure brand perception objectively and continuously rather than through expensive surveys?
- Q 4** How can we distinguish sentiment between individual product lines (iPhone, iPad, Android, Google apps)?
- Q 5** How do we prevent class imbalance (only 6% negative) from hiding the most critical business signals?

Data Understanding

Dataset

CrowdFlower via data.world
9,093 human-labelled tweets
Columns: tweet_text · product · sentiment

9,093

Raw tweets

8,892

After cleaning

60%

Neutral class

6%

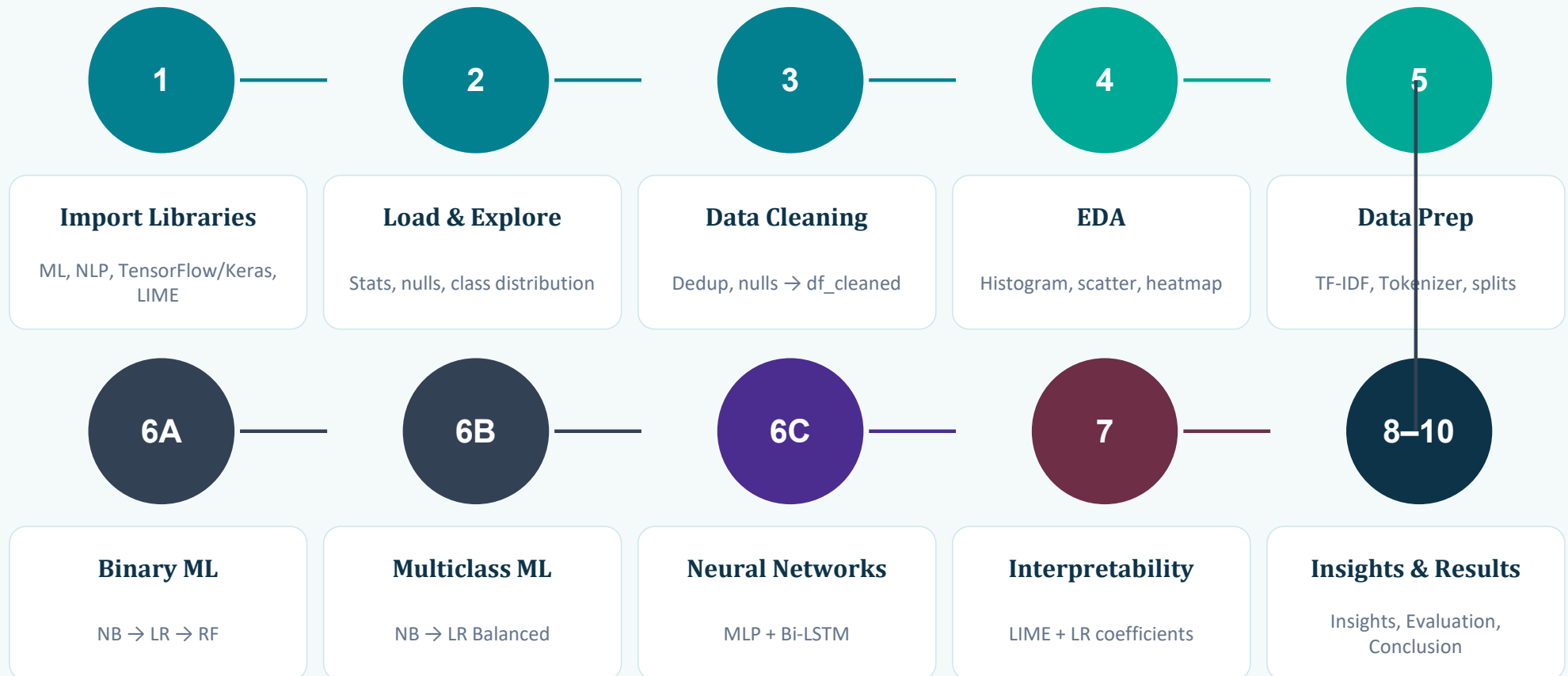
Negative (critical)

Sentiment Distribution



Step2: Load, Explore Data&Project Structure

10-step systematic notebook — three modelling phases flow sequentially before interpretability.



Step 3: Data Cleaning → df_cleaned

1

Drop helper column

Remove tweet_length
created during EDA

2

Remove duplicates

22 duplicate
rows deleted

3

Drop null tweets

1 row with missing
tweet_text removed

4

Remove ambiguous

156 'Can't Tell'
labels dropped

5

Save df_cleaned

8,892 rows × 3 cols
index reset

Before vs After Cleaning

	Rows	Duplicates	Null tweets	Ambiguous labels
Before	9,093	22	1	156
After	8,892	0	0	0

Step 5: Data Preparation / Preprocessing

NLP Text Cleaning Pipeline

- 1 Remove URLs, @mentions, #hashtags
- 2 Lowercase all text
- 3 Strip punctuation and digits
- 4 Remove NLTK stop-words
- 5 Lemmatize (WordNetLemmatizer)
- 6 TF-IDF — unigrams+bigrams, max 5,000 features

Encoding — All Columns Numerical

tweet_text	Text	TF-IDF / Keras Tokenizer
product	Categorical	LabelEncoder
sentiment	Categorical	LabelEncoder

Two Feature Formats Prepared

TF-IDF matrix (5,000 features)

Used by: NB, LR, RF, MLP

Keras integer sequences (max len 80)

Used by: Bidirectional LSTM

Stratified 80/20 split — same for ALL models

Steps 6A & 6B: Traditional ML Models

Phase A — Binary (Positive vs Negative)

Naïve Bayes

Baseline

Acc 84.6% F1 0.916

Fast baseline; treats each word independently

Logistic Regression

Iteration 2

Acc 85.5% F1 0.920

Bigrams + regularisation; marginal gain

Random Forest

★ Best Binary

Acc 88.6% F1 0.936

Ensemble; best accuracy & F1

Phase B — Multiclass (Positive / Neutral / Negative)

Naïve Bayes — Baseline

Accuracy: 66.3% Weighted F1: 0.606 Neg. Recall: 11%

Biased toward majority Neutral class

LR Balanced — Best Traditional

Accuracy: 63.8% Weighted F1: 0.649 Neg. Recall: 58%

`class_weight='balanced'` — critical for Negative recall

Step 6C: Neural Network Models (Phase C)

Applied after traditional ML — same train/test split for fair comparison.

MLP Classifier — scikit-learn

Neural Network on TF-IDF features

Architecture

TF-IDF(5000) → Dense(128, ReLU) → Dense(64, ReLU) → Softmax(3)

Accuracy	67.5%
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Weighted F1	0.657
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vs LR Balanced	+0.010 F1
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Early stopping	✓ Applied
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★ **Best Multiclass Model — Highest Weighted F1**

Bidirectional LSTM — TensorFlow/Keras

Deep Learning — reads tweet word-by-word

Architecture

Embedding(10k,64) → BiLSTM(64) → MaxPool → Dense(64) → Softmax(3)

Accuracy	67.1%
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Weighted F1	0.638
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Epochs (early stop)	5 of 15
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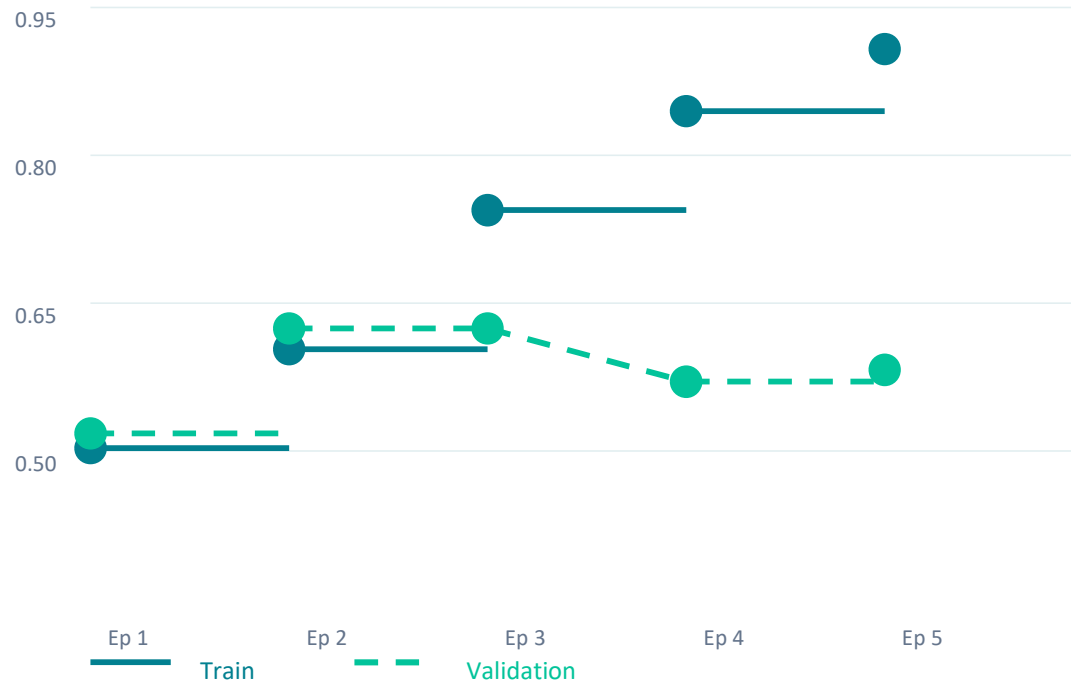
Early stopping	✓ Applied
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Reads forward AND backward — captures word order unlike TF-IDF

Step 6C: LSTM Training History

Bidirectional LSTM — 5 epochs (early stopping). Validation peaked at epoch 2; best weights restored.

Accuracy per Epoch



Key Observations

- 1 Train acc rises 60%→87% across 5 epochs
- 2 Validation peaks at epoch 2 (68.3%) then stabilises
- 3 Early stopping fired — best weights from epoch 2 restored
- 4 Train/val gap: more data would improve generalisability

All Models — Performance Comparison

Multiclass (3-class) results across all four models evaluated on the same stratified test set.

Naïve Bayes Baseline

Accuracy **66.3%**

Weighted F1 **0.606**

Neg. Recall **11%**

sklearn

LR Balanced Trad. Best

Accuracy **63.8%**

Weighted F1 **0.649**

Neg. Recall **58%**

sklearn

MLP ★ Best Overall

Accuracy **67.5%**

Weighted F1 **0.657**

Neg. Recall **~55%**

sklearn

Bi-LSTM Deep Learning

Accuracy **67.1%**

Weighted F1 **0.638**

Neg. Recall **0%**

Keras

Weighted F1 at a glance:

NB 0.606

LR 0.649

MLP ★ 0.657

LSTM 0.638

Step 7: Model Interpretability — LIME

Comes after all three modelling phases — explains decisions from the best models.

LIME — Local + Global Interpretability

- 1 Takes a single tweet to explain
- 2 Perturbs text by hiding words randomly
- 3 Observes how the model prediction changes
- 4 Ranks words by influence on the result
- 5 Outputs which words drove the prediction

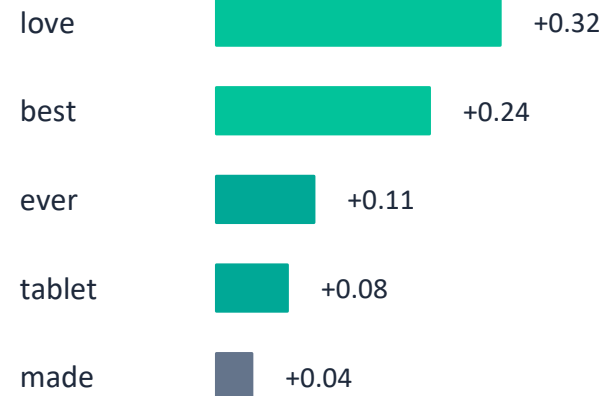
Global view: LR coefficient chart shows which words the model associates with each class across ALL tweets.

Example Prediction Explained

"Love the new iPad — best tablet ever!"

Predicted: POSITIVE (confidence 0.92)

Words driving prediction:



Top Positive: love · great · awesome · new ipad

Top Negative: fail · crash · problem · broken

Step 8: Key Insights

1

6% negative tweets — severe imbalance; weighted F1 over accuracy

2

Negative tweets longer (109 chars) — users elaborate when frustrated

3

iPad & Apple drive most positive sentiment; Google leans neutral

4

Bigrams improve binary accuracy 84.6%→88.6% — phrases matter

5

MLP best multiclass F1 (0.657) — non-linear TF-IDF combinations

6

Bi-LSTM competitive (67.1%) — more data needed to exceed ML

7

Early stopping: LSTM stopped at 5/15 epochs, overfitting prevented

8

LIME confirms: 'love','great' → Positive; 'fail','crash' → Negative

9

`class_weight='balanced'` raises Negative recall 11%→58%

10

Sarcasm unsolved — BERT is the recommended next modelling step

Step 9: Evaluation

88.6%

Binary Accuracy
Random Forest

0.936

Binary F1
Positive class

67.5%

Best Multiclass
MLP Classifier

0.657

Best Multiclass
Weighted F1

Final Model Summary

Model	Acc	W-F1
Random Forest	88.6%	0.936
LR Balanced	63.8%	0.649
MLP ★	67.5%	0.657
Bi-LSTM	67.1%	0.638

Validation Approach

- Stratified 80/20 train/test split — same for all models
- LSTM: extra 10% validation for early stopping
- Pipeline prevents TF-IDF data leakage
- Weighted F1 over accuracy — 60% = just predicting Neutral
- Negative recall tracked separately — highest business cost

Conclusion, Recommendations & Next Steps

Deploy Binary RF Model

88.6% accuracy — ready for real-time positive/negative tweet alerting today.

Deploy MLP for Multiclass

Best weighted F1 (0.657) — suitable for brand sentiment dashboards.

Collect More Negatives

Only 6% negative tweets. More labelled data will significantly improve recall.

Fine-tune BERT Next

Transformer models handle sarcasm and context that TF-IDF and LSTM cannot.

Next Steps:

GridSearchCV tuning • SMOTE oversampling • Twitter API pipeline • Time-series tracking • Fine-tune BERT

Thank You

Twitter Sentiment Analysis — Phase 4 NLP Project

Dataset

9,093 tweets
CrowdFlower

Best Binary

Random Forest
88.6% / F1=0.936

Best Multiclass

MLP Classifier
67.5% / F1=0.657

Deep Learning

Bi-LSTM
67.1% / F1=0.638

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Questions?